BIOLOGY · CLASS 11

Chapter 7: Photosynthesis in Higher Plants

7.1 What is Photosynthesis?

Photosynthesis is the process by which green plants, algae and some bacteria convert light energy into chemical energy. Using sunlight, water and carbon dioxide, plants make glucose and release oxygen as a by-product. The overall equation is 6CO2 + 6H2O + light energy → C6H12O6 + 6O2. Almost every food chain on Earth depends on this process, because plants are the primary producers that capture energy from the Sun.

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7.2 Where Does It Happen?

Photosynthesis takes place in the chloroplasts, found mainly in the mesophyll cells of leaves. Each chloroplast has a double membrane, a fluid called the stroma, and stacks of flattened sacs called thylakoids, arranged in grana. Chlorophyll a and chlorophyll b, located in the thylakoid membranes, absorb mostly blue and red light and reflect green light, which is why leaves look green.

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7.3 The Light Reactions

The light-dependent reactions happen in the thylakoid membranes. Light energy excites electrons in Photosystem II and Photosystem I. Water is split (photolysis) to replace lost electrons, releasing oxygen. The flow of electrons through the electron transport chain pumps protons into the thylakoid space, and ATP synthase uses this gradient to make ATP. NADP+ is reduced to NADPH. ATP and NADPH carry energy to the next stage.

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7.4 The Calvin Cycle

The light-independent reactions, or Calvin cycle, take place in the stroma. The enzyme RuBisCO fixes carbon dioxide onto ribulose bisphosphate (RuBP). Using ATP and NADPH from the light reactions, the products are reduced to glyceraldehyde-3-phosphate (G3P). Some G3P leaves the cycle to build glucose, sucrose and starch; the rest regenerates RuBP so the cycle can continue. Three turns of the cycle fix three molecules of CO2.

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7.5 C3, C4 and CAM Plants

Most plants are C3 plants, where the first stable product is a three-carbon compound. In hot, dry climates RuBisCO can bind oxygen instead of CO2, a wasteful process called photorespiration. C4 plants such as maize and sugarcane first fix CO2 into a four-carbon compound in mesophyll cells and pass it to bundle-sheath cells, keeping CO2 levels high around RuBisCO. CAM plants such as cactus and pineapple open their stomata at night and store CO2 as malic acid to save water.

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7.6 Factors Affecting the Rate

The rate of photosynthesis depends on light intensity, carbon dioxide concentration, temperature and water availability. According to Blackman's law of limiting factors, when a process depends on several factors, its rate is limited by the factor that is in shortest supply. Increasing light raises the rate only until another factor, such as CO2 or temperature, becomes limiting. Very high temperatures denature enzymes and reduce the rate.

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7.7 Importance of Photosynthesis

Photosynthesis provides food for nearly all living organisms, maintains the oxygen level of the atmosphere, and removes carbon dioxide, helping to regulate the climate. Fossil fuels such as coal and petroleum are stored products of ancient photosynthesis. Understanding photosynthesis also helps scientists improve crop yields and develop clean energy technologies such as artificial leaves.

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Chapter Summary

Photosynthesis converts light energy into chemical energy in chloroplasts. The light reactions in the thylakoids produce ATP, NADPH and oxygen. The Calvin cycle in the stroma uses them to fix CO2 into sugars. C4 and CAM plants reduce photorespiration in hot or dry conditions. The rate is controlled by the limiting factor. Exercises: (1) Write the overall equation of photosynthesis. (2) Differentiate between the light reactions and the Calvin cycle. (3) Explain photorespiration. (4) Compare C3 and C4 plants. (5) State Blackman's law of limiting factors.

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